

Orthographic-to-Semantic Abstraction in Matched Spiking and Artificial Recurrent Networks

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Abstract

How small spiking encoders internally transform linguistic form into meaning remains poorly understood. We study a three-layer recurrent spiking neural network (SNN; 18,128 parameters) trained from one-hot grapheme events with a semantic triplet objective, without subword tokenization, pretrained embeddings, attention, or a Transformer backbone. A parameter-matched artificial recurrent neural network (ANN) serves as a control. In a curated factorial diagnostic, the SNN changes from orthographic to semantic organization: an abstraction index rises from -0.854 in layer 1 to $+0.188$ in layer 3 (95% CI $[0.075, 0.304]$). This shift depends strongly on input order: permutation reduces rate-readout semantic triplet accuracy (STA) by 0.259. In contrast, destroying precise output timing has little effect (original van Rossum STA = 0.910; rate-only = 0.913). Thus, temporal dynamics help *construct* the representation, but precise spike timing is not required to *express* its final semantic organization. The ANN is numerically higher on IID evaluation and generalizes better to held-out relations and context-exposed lexemes, whereas the SNN is more robust to orthographically deceptive negatives and operationally sparser. The hierarchy is therefore not unique to spikes, but matched spiking and non-spiking encoders exhibit distinct representational profiles. These results motivate a distinction between temporal computation and temporal representation in spiking language systems.

1 Introduction

Spiking neural networks (SNNs) have recently scaled from spike-based document representations [Białas et al., 2020] to generative and general-purpose language models [Zhu et al., 2023, Xing et al., 2024, Tang, 2026]. Most work emphasizes architecture, task accuracy, or the prospect of sparse event-driven computation. We ask a complementary question: *how does linguistic organization change inside a small recurrent SNN that receives only elementary graphemic events?*

Direct processing of bytes or characters can support competitive language models and improve robustness on spelling-sensitive tasks [Xue et al., 2022]. It also creates a useful setting for representational analysis: semantic structure cannot enter through pretrained lexical embeddings.

Our central questions are whether semantic organization emerges with depth, what role temporal order and spike timing play, and whether the resulting hierarchy is specific to spikes.

We make three contributions. First, a curated orthography $\times$ semantics diagnostic reveals a layerwise transition from form-dominated to meaning-dominated geometry. Second, input perturbations and output readouts dissociate temporal *computation* from temporal *representation*: sequence order is important, but precise final spike timing is not. Third, comparison with an exactly parameter-matched ANN shows that the hierarchical transition is shared, while the models differ in hard-negative robustness, sequence dependence, sparsity, and generalization.

2 Related Work

SpikeGPT demonstrated sequential language generation with large backpropagation-trained SNNs [Zhu et al., 2023]; SpikeLM introduced elastic bi-spiking mechanisms for discriminative and generative language tasks [Xing et al., 2024]; and NeuronSpark reported end-to-end training of a 0.9B-parameter spiking language model from random initialization [Tang, 2026]. Earlier work learned low-dimensional spike-based document representations with STDP [Białas et al., 2020], while other text classifiers first convert pretrained word embeddings into spike trains [Lv et al., 2023]. Our setting is intentionally smaller and diagnostic: semantic supervision is applied directly to recurrent encoders receiving one-hot graphemes.

Adaptive spiking neurons can expand the memory and computational capacity of recurrent SNNs [Bellec et al., 2018], and surrogate gradients enable optimization despite nondifferentiable spikes [Neftci et al., 2019]. To examine the resulting code, we use the van Rossum distance, whose time constant interpolates between sensitivity to temporal coincidence and firing-rate differences [van Rossum, 2001]. A recent MEG study also reported earlier orthographic than semantic information during natural reading [Wang et al., 2025]; this is an informative comparison, not a claim that our model is neurobiologically equivalent.

3 Method

3.1 Models and input

The primary model is a three-layer recurrent SNN with adaptive leaky integrate-and-fire (ALIF) neurons. Decay parameters β are heterogeneous and learnable. Each grapheme is presented as a one-hot event; the model uses neither BPE nor pretrained semantic embeddings. A post-stimulus interval allows recurrent activity to evolve after the input ends. Training uses surrogate gradients, an activity/sparsity penalty, and a semantic triplet loss. All experiments ran on CPU.

The control ANN was designed to contain exactly 18,128 trainable parameters, matching the SNN. It preserves the same input, depth, width, recurrence, adaptation, decay initialization, seeds, data, and contrastive objective, while replacing binary spikes with adaptive leaky-tanh states. The comparison therefore isolates representational profile more closely than comparisons against a large pretrained language model, although neuron dynamics necessarily remain different.

3.2 Distances and evaluation

For anchor a , semantic positive p , and negative n , semantic triplet accuracy is

$$\text{STA} = \frac{1}{N} \sum_{i=1}^N \mathbf{1}[d(a_i, p_i) < d(a_i, n_i)]. \quad (1)$$

Spike trains are compared with van Rossum distance [van Rossum, 2001]; a rate-only readout discards within-window spike timing. Negatives are either random or *hard orthographic* distractors that resemble the anchor in form but not meaning.

To separate form and meaning, a curated diagnostic contains 48 pairs (12 per cell) in a 2×2 orthography $\times$ semantics design. At each layer l , predefined semantic and orthographic distance contrasts, S_l and O_l , yield

$$A_l = S_l - O_l, \quad (2)$$

where $A_l < 0$ denotes orthographic dominance and $A_l > 0$ semantic dominance. We additionally test relation-disjoint generalization, removing each evaluated positive relation in both directions from training, and supervised contextual transfer to eight new lexemes.

3.3 Temporal and dynamical controls

Output controls apply local or global spike-time shuffling while preserving each neuron’s spike count, as well as a rate-only readout. Input controls reverse or permute graphemes, insert random gaps, or replace order with a bag-of-graphemes representation. Timescale controls compare a single fixed β , fixed layerwise β , fixed heterogeneous β , and learnable heterogeneous β . Fixed-decay variants have 17,968 trainable parameters and the learnable variant

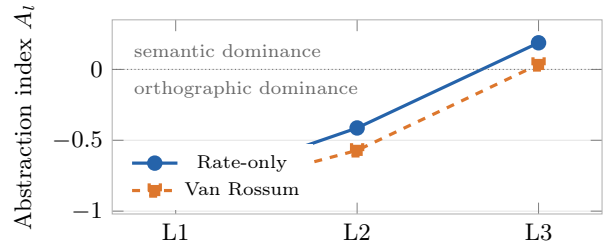


Figure 1: Hierarchical form-to-meaning shift in a curated diagnostic. The orthography–semantics abstraction index crosses zero only in the deepest layer. The layer-3 rate-only estimate has a seed-level 95% CI of $[0.075, 0.304]$; lexical pairs are fixed.

18,128, while all retain 18,128 state values. Principal reproducibility experiments use seeds 7, 19, 31, 43, and 59, a fixed set of 260 validation triplets, and repeated corruptions. Reported percentile-bootstrap intervals use seeds as the statistical unit; paired intervals are used for within-seed comparisons. Because lexical items are fixed, these intervals do not quantify uncertainty over alternative lexical samples.

4 Results

4.1 Depth reverses orthographic dominance

Within the curated factorial diagnostic, the rate-only abstraction index changes monotonically from $A_1 = -0.854$ to $A_2 = -0.413$ and $A_3 = +0.188$ (95% CI $[0.075, 0.304]$; Fig. 1). Van Rossum readouts show the same qualitative transition $(-0.911, -0.571, +0.035)$. At layer 3, graphically dissimilar synonyms are slightly closer on average than graphically similar semantic mismatches (0.627 vs. 0.661 van Rossum distance). This is an internal diagnostic on fixed lexical items, not a causal factorial experiment or evidence that the effect generalizes to the wider lexicon.

Triplet geometry corroborates the factorial result. Overall STA increases from 0.632 (L1) to 0.768 (L2) and 0.910 (L3), while the separation margin grows from 0.009 to 0.797. Hard orthographic negatives are initially closer than semantic positives, producing negative margins in L1 (-0.049) and L2 (-0.098). L3 reverses the ordering (margin = $+0.156$, STA = 0.640). Depth thus expands the geometry while selectively correcting a form-based shortcut; the result is a relative transformation rather than absolute convergence.

Post-stimulus activity also increases with depth (0.001, 0.021, and 0.045 for L1–L3), consistent with greater recurrent persistence. This alone does not establish conceptual attractors.

4.2 Temporal computation without precise temporal coding

Output-time perturbations provide little support for a precise temporal semantic code. Original van Rossum

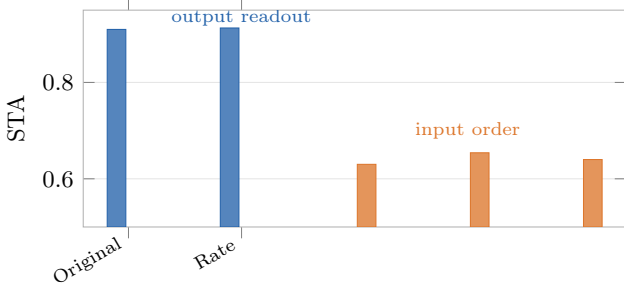


Figure 2: Temporal computation versus representation. Discarding output timing (rate-only) preserves STA, whereas disrupting input order produces large losses. Values use the corresponding primary readout: van Rossum for original and rate-only otherwise.

STA is 0.910 (95% CI [0.876, 0.935]); local shuffling is unchanged, global shuffling gives 0.886, and rate-only reaches 0.913 (95% CI [0.866, 0.945]). Similarly, the best single slow van Rossum scale ($\tau = 14$, STA = 0.914) is statistically indistinguishable from rate-only ($\Delta = 0.001$, CI [-0.012, 0.016]), while combining scales provides no benefit.

Input order, however, is essential (Fig. 2). Rate-only STA falls from 0.913 to 0.630 under reversal, 0.654 under permutation, and 0.640 for bag-of-graphemes input. Mild and strong gap jitter cause smaller drops to 0.867 and 0.824. On hard distractors, the positive L3 margin under correct order (+0.156) becomes negative under permutation (-0.109) or bagging (-0.129). The key distinction is therefore

ordered input $\rightarrow$ recurrent dynamics $\rightarrow$ population code,

not preservation of exact spike times as the semantic output format.

4.3 Heterogeneous neuronal dynamics

Under the van Rossum readout used for the decay ablation, learnable heterogeneous decays achieve overall STA = 0.910 and hard-negative STA = 0.640, compared with 0.892 and 0.490 for a single fixed β . The overall difference is inconclusive ($\Delta = 0.018$, CI [-0.032, 0.065]), but the hard-negative gain is reliable ($\Delta = 0.150$, CI [0.020, 0.280]), as is the L3 hard-margin gain (0.127, CI [0.016, 0.226]). The later population comparison uses rate-only SNN STA, explaining its higher hard-negative value of 0.720.

This supports heterogeneous dynamics, but not a strong claim for learning the timescales: fixed heterogeneity reaches hard STA = 0.520, and learned β values move only modestly. Mean $|\Delta\beta|$ decreases from 0.0127 in L1 to 0.0055 in L3, with no strong L3 correlation between β and post-stimulus or hard-example firing. The parsimonious interpretation is that initial heterogeneity supplies much of the benefit and learning makes small adjustments.

Table 1: Matched 18,128-parameter model comparison. Bold marks the higher value, except that permutation drop is descriptive rather than “higher is better.”

Metric	SNN	ANN
IID STA	0.913	0.928
Hard orthographic STA	0.720	0.540
Factorial A_3	0.188	0.141
Permutation drop	0.259	0.133
Relation-disjoint	0.778	0.906
Contextual transfer	0.758	0.833

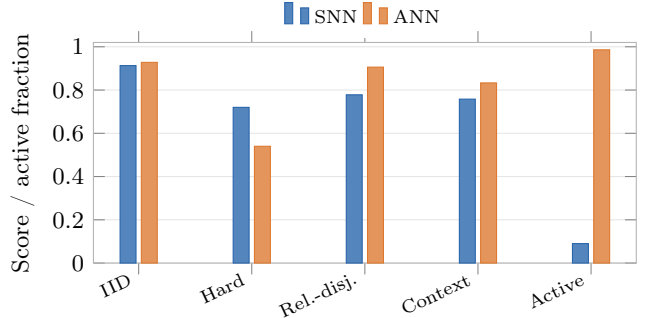


Figure 3: Distinct matched-model profiles. The ANN generalizes better, while the SNN better rejects deceptive orthographic negatives and has a lower active fraction. “Active” denotes spike rate for the SNN and $P(|\text{state}| > 10^{-3})$ for the ANN, so it is an operational, not energetic, comparison.

4.4 Matched SNN–ANN comparison

The matched ANN is numerically higher on IID STA (0.928 vs. 0.913), although the paired difference is inconclusive (-0.015 , CI [-0.056, 0.012]). It generalizes better on relation-disjoint evaluation (0.906 vs. 0.778; paired difference = -0.128 , CI [-0.200, -0.056]) and contextual transfer (0.833 vs. 0.758; paired difference = -0.075 , CI [-0.142, -0.008]; Table 1). Using population/integrated-state readouts, the SNN is better on hard orthographic negatives (0.720 vs. 0.540; difference = 0.180, CI [0.040, 0.360]) and shows a larger permutation drop (0.259 vs. 0.133).

The SNN spike rate is 0.090, whereas 98.6% of ANN states exceed 10^{-3} in magnitude. These quantities are not physically identical, but they establish a much sparser *operational state* in the SNN. They do not establish lower energy use, which would require a hardware- and operation-aware measurement.

4.5 Generalization and contextual transfer

When evaluated semantic relations are excluded from training in both directions, both models remain above chance: SNN STA is 0.778 (CI [0.706, 0.844]) and ANN STA is 0.906 (CI [0.850, 0.961]). Thus the encoders do more than memorize the exact evaluated pairs, although the ANN generalizes substantially better.

For eight new lexemes, isolated-word STA begins near chance (SNN 0.487; ANN 0.546). After supervised expo-

sure inside sentences, it rises to 0.758 for the SNN (gain = 0.271, CI [0.208, 0.333]) and 0.833 for the ANN (gain = 0.288, CI [0.254, 0.313]). This is supervised contextual transfer: contextual evidence can shape a representation later used in isolation, but the experiment does not demonstrate unsupervised semantic acquisition.

5 Discussion

The strongest result is a dissociation. Temporal structure matters greatly on the way *into* the recurrent hierarchy, yet precise timing matters little in the final semantic readout. The network appears to use sequence-sensitive dynamics to build a stable population-level organization. Calling the result either purely temporal or purely rate-coded would miss this computational distinction.

The form-to-meaning transition itself is not specific to spiking neurons: the matched ANN also develops it. The contribution of spikes is instead a different profile. Relative to the ANN, the SNN is operationally sparser, more dependent on sequence order, and more robust to orthographic shortcuts. Its IID score is numerically lower but not conclusively different, while its relation-disjoint and contextual generalization are reliably lower. The hard-negative advantage suggests that recurrent spiking dynamics may impose a useful inductive bias against superficial similarity, but the present experiments do not isolate its exact mechanism.

Heterogeneous neuronal timescales partially contribute to this profile. Replacing them with one fixed decay mainly harms the difficult orthographic condition, while the global score changes little. Since fixed heterogeneity remains competitive and learned decays move only slightly, diversity of dynamics is better supported than strong timescale specialization.

Limitations. The dataset is small and Spanish-only, and the factorial diagnostic contains 12 pairs per cell. Its cells differ in semantic relation type and supervision exposure, lexical frequency is uncontrolled, and lexical selection may influence the contrasts. Seed-level confidence intervals do not represent uncertainty over alternative lexical samples. Semantic training and contextual transfer are supervised. Both networks are small and were evaluated on CPU. Sparse activity is not equivalent to energy efficiency, and no neuromorphic hardware measurement was made. The model receives graphemes rather than audio, and the findings do not establish speech understanding, biological plausibility, or conceptual attractors. Broader corpora, preregistered lexical controls, additional matched recurrent baselines, and direct hardware measurements are needed.

6 Conclusion

A recurrent hierarchy receiving ordered graphemic events can progressively transform orthographic organization into

semantic organization. In the SNN, input order is important for constructing that transformation, while heterogeneous dynamics contribute specifically to hard-negative separation relative to a single fixed decay. The final semantic geometry is captured predominantly by a sparse population/rate code rather than precise spike timing. A matched ANN shares the hierarchy and generalizes better on relation-disjoint and contextual tests, whereas the SNN better resists orthographically misleading negatives. The natural next step is to test whether richer temporal signals, including speech, converge toward a shared conceptual population geometry.

Code availability. Source code, experiment scripts, and reproducibility instructions are hosted at <https://github.com/Danval-003/snn-semantic-dynamics>.

References

- Guillaume Bellec, Darjan Salaj, Anand Subramoney, Robert Legenstein, and Wolfgang Maass. Long short-term memory and learning-to-learn in networks of spiking neurons. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Marcin Białas, Marcin Michał Mironczuk, and Jacek Mańdziuk. Biologically plausible learning of text representation with spiking neural networks. In *Parallel Problem Solving from Nature-PPSN XVI*, pages 433–447. Springer, 2020. doi: 10.1007/978-3-030-58112-1_30.
- Changze Lv, Jianhan Xu, and Xiaoqing Zheng. Spiking convolutional neural networks for text classification. In *International Conference on Learning Representations*, 2023.
- Emre O. Neftci, Hesham Mostafa, and Friedemann Zenke. Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36(6):51–63, 2019. doi: 10.1109/MSP.2019.2931595.
- Zhengzheng Tang. NeuronSpark: A spiking neural network language model with selective state space dynamics. *arXiv preprint arXiv:2603.16148*, 2026.
- M. C. W. van Rossum. A novel spike distance. *Neural Computation*, 13(4):751–763, 2001. doi: 10.1162/089976601300014321.
- Lijuan Wang, Steven Frisson, Yali Pan, and Ole Jensen. Fast hierarchical processing of orthographic and semantic parafoveal information during natural reading. *Nature Communications*, 16:8893, 2025. doi: 10.1038/s41467-025-63916-y.
- Xingrun Xing, Zheng Zhang, Ziyi Ni, Shitao Xiao, Yiming Ju, Siqi Fan, Yequan Wang, Jiajun Zhang, and Guoqi Li. SpikeLM: Towards general spike-driven language

modeling via elastic bi-spiking mechanisms. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 54698–54714. PMLR, 2024.

Linting Xue, Aditya Barua, Noah Constant, Rami Al-Rfou, Sharan Narang, Mihir Kale, Adam Roberts, and Colin Raffel. ByT5: Towards a token-free future with pre-trained byte-to-byte models. *Transactions of the Association for Computational Linguistics*, 10:291–306, 2022. doi: 10.1162/tacl_a_00461.

Rui-Jie Zhu, Qihang Zhao, Guoqi Li, and Jason K. Eshraghian. SpikeGPT: Generative pre-trained language model with spiking neural networks. *arXiv preprint arXiv:2302.13939*, 2023.